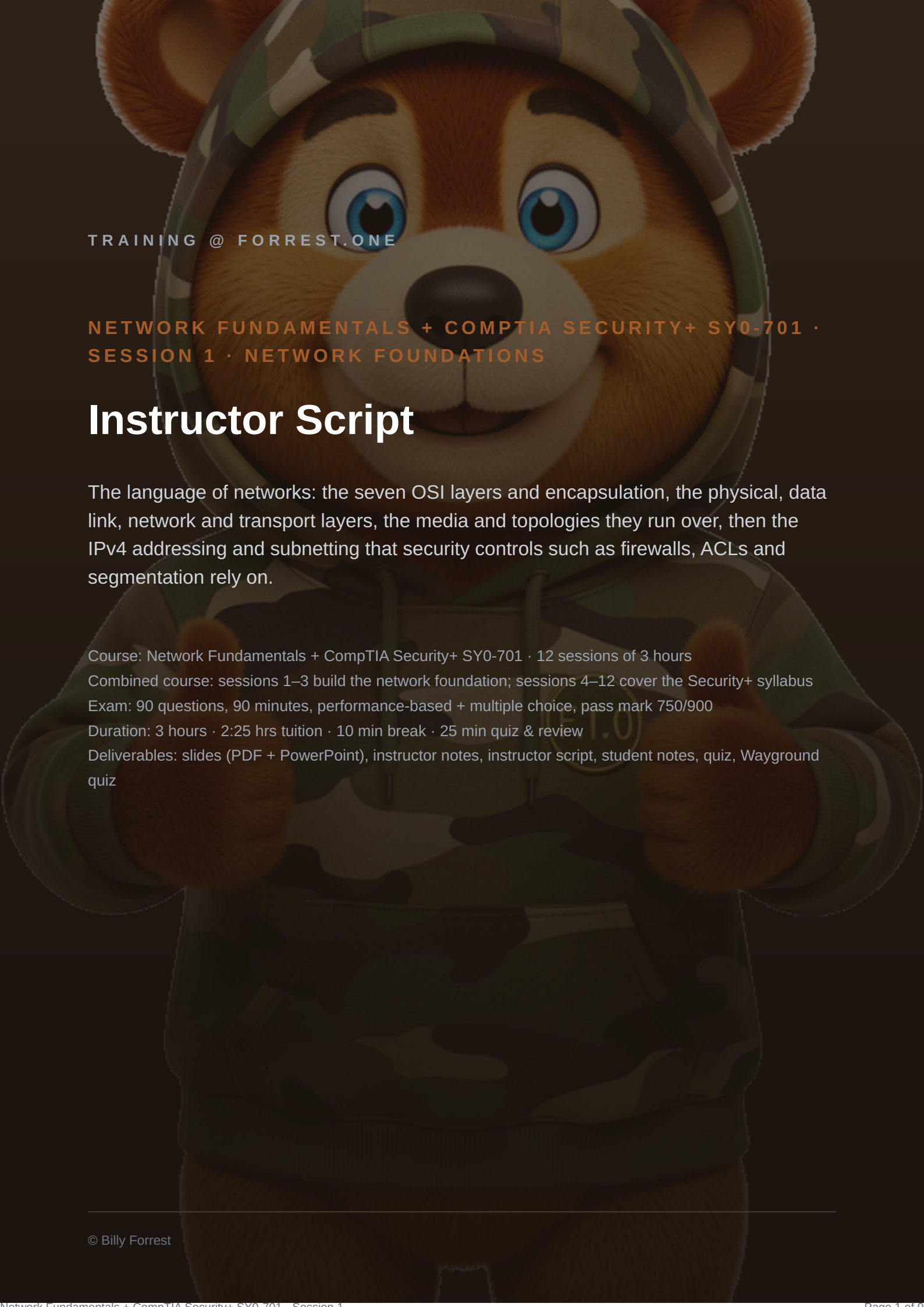




TRAINING @ FORREST.ONE

NETWORK FUNDAMENTALS + COMPTIA SECURITY+ SY0-701 ·
SESSION 1 · NETWORK FOUNDATIONS

Instructor Script

The language of networks: the seven OSI layers and encapsulation, the physical, data link, network and transport layers, the media and topologies they run over, then the IPv4 addressing and subnetting that security controls such as firewalls, ACLs and segmentation rely on.

Course: Network Fundamentals + CompTIA Security+ SY0-701 · 12 sessions of 3 hours

Combined course: sessions 1–3 build the network foundation; sessions 4–12 cover the Security+ syllabus

Exam: 90 questions, 90 minutes, performance-based + multiple choice, pass mark 750/900

Duration: 3 hours · 2:25 hrs tuition · 10 min break · 25 min quiz & review

Deliverables: slides (PDF + PowerPoint), instructor notes, instructor script, student notes, quiz, Wayground quiz

Instructor Script — Network Foundations: Networking Models & IP Addressing

A word-for-word delivery script for Session 1. Stage directions appear in square brackets. Each slide runs roughly 45 to 90 seconds of speech; activities and discussions expand the timings given in the instructor notes.

Slide 1 · Title

Good day, and welcome to the combined Network Fundamentals and CompTIA Security+ course: twelve sessions of three hours. The first three sessions build your network foundation. How data moves across a network, how signals travel, and how devices are addressed. Everything after that, the threats, the cryptography, the firewalls and the operations, assumes you can read a network like a map. This session is where you learn the legend. By the end of today you will be able to walk a message down the stack, choose the right cable for a job, and read a subnet mask.

Slide 2 · The journey

Here is the shape of the course. Sessions one to three, the network foundation: models, addressing, switching and services. Sessions four to seven, the security core: fundamental concepts, threats, cryptography and identity. Sessions eight and nine, security architecture: network, endpoint, application and cloud. Sessions ten to twelve, operations and governance: vulnerability management, incident response, and risk and compliance. And the exam you are building towards: ninety questions, ninety minutes, performance-based questions alongside multiple choice, pass mark seven hundred and fifty out of nine hundred. Today is session one of that journey, and it is split into six chapters: the language of networks, the OSI model, physical and data link, network and transport, media and topologies, then IPv4 addressing and subnetting.

Slide 3 · Session objectives

By the end of these three hours you will be able to do ten things. Name all seven OSI layers and each one's job. Explain encapsulation, which is how data gains a header at every layer. Describe the physical layer, including its signals, media and connectors. Explain Media Access Control addressing on a local segment. Compare Transmission Control Protocol with User Datagram Protocol. Describe the upper layers: session, presentation and application. Choose suitable wired and wireless media for a given job. Compare star, mesh and hybrid topologies. Read an IPv4 address with its subnet mask. And calculate subnets using classless addressing. Ten objectives, and the quiz at the end of the session tests every one of them.

Slide 4 · Why layers matter at all

Before any layer names, let us establish why we bother. Picture troubleshooting with no structure at all. A user cannot reach a website, and you have no idea whether the fault is the cable, the switch, the route or the server. That is impossible work. Now split the problem into seven layers, each with one responsibility. You test them in turn and you eliminate as you go. That is the whole value of the model, and it is why I will come back to layer thinking all day. One more thing: this model is a map, not a product. Nobody sells you a layer five. The value is in the thinking.

Slide 5 · The seven layers

Here are the seven layers, counted down from the top. Layer seven, application: the services users touch, such as web and email. Layer six, presentation: encoding, compression and encryption. Layer five, session: opening, managing and closing conversations. Layer four, transport: delivering end to end, and using port numbers. Layer three, network: logical addressing and routing. Layer two, data link: delivering frames on a local segment. Layer one, physical: bits on the wire, light in the fibre, or radio waves. Learn the names, the order, and one protocol or device for each.

Slide 6 · The journey of a message

This is the slide that makes everything else click, so follow one message all the way down. The user types into an application, so the data starts at the top, layers seven to five, and nothing has been added yet. Layer four splits it into segments and adds port numbers, so the far machine knows which application should receive it. Layer three adds a logical address, an IP address, so the data can be routed between networks. Layer two adds a hardware address, so it reaches the right device on the local segment. Layer one turns the whole thing into signals. And the receiver reverses the process, peeling one header off at a time on the way back up.

Slide 7 · Encapsulation

That wrapping process has a name, and it is encapsulation. Each layer adds its own header as the data travels down, and what changes is what we call the unit. At the transport layer it is a segment for Transmission Control Protocol, or a datagram for User Datagram Protocol. At the network layer it becomes a packet. At the data link layer it is a frame. And at the physical layer it is simply bits on the medium. Same data, five names. Know this vocabulary precisely, because exam questions use these words exactly and they will not accept a near miss.

Slide 8 · The physical layer

Now the bottom two layers, the ones you can physically touch, and the place where most total outages live. Layer one carries the raw bitstream between devices, with no understanding of meaning at all. It supports three signal types. Copper carries electrical signals, using voltage changes to represent ones and zeros. Fibre carries light pulses, which travel much further with far less interference. Wireless carries radio waves

through the air, which buys you mobility. This layer also defines the standards: the cabling, the connectors and the signalling. And remember the first rule of troubleshooting. If layer one fails, nothing above it can possibly work.

Slide 9 · Signals and their media

This table keeps the three signal types straight, and it is really a table of trade-offs. Electrical signals run over copper, which is cheap and widely installed. Light runs through fibre, which spans the longest distances and is immune to electrical interference. Radio travels through the air, which gives you mobility at the cost of sharing the channel with everyone else. Notice that every medium trades cost against distance and interference. Those three factors will guide almost every cabling decision you make in your career, so hold on to them; we will use them again at the end of this chapter.

Slide 10 · The data link layer

Layer two handles node to node delivery on the same local network. Its addressing scheme is the Media Access Control address: forty eight bits long, written in hexadecimal, something like zero zero one A two B three C four D five E. The unit of data here is called a frame. The layer splits into two sublayers, and CompTIA expects you to know both, which is the next slide. It also appends a Cyclic Redundancy Check, a mathematical checksum used to detect corruption in transit. And here is the security point: these addresses are assigned by the manufacturer, but they can be changed in software. So a hardware address is not a trustworthy identity, and you should never treat it as proof of who is on your network.

Slide 11 · The two data link sublayers

The two sublayers divide the work, and the direction each one faces is the key to remembering them. Logical Link Control sits closer to the network layer. It works out which layer three protocol is inside the frame, such as IP, and it handles error checking and frame synchronisation. Media Access Control sits closer to the physical layer. It decides how devices share the medium without colliding, and it delivers the frame using the hardware address. So Logical Link Control looks up, and Media Access Control looks down. That is exactly how exam questions phrase it.

Slide 12 · A dead link is almost always layer 1

Here is a habit I want you to build before we go any further. A user reports they have lost connectivity completely. The temptation is to dive into routing tables or firewall rules, because that feels clever and it feels like real network engineering. Resist it. Check the cable, the port indicator and the link state first. A completely dead link is usually physical, and physical faults are both the most common and the cheapest to repair. Only once layer one is sound does it make sense to look higher up the stack.

Slide 13 · The network layer

Layer three is where logical addressing and routing live, and the distinction that matters most is this. A hardware address identifies a device on a local segment. An IP address identifies a device across networks. The network layer moves packets between networks, and routers are the devices that operate here. Every packet carries a Time to Live value that drops by one at each router, and when it reaches zero the packet is discarded, which stops lost packets circulating forever. Data crosses the internet one router at a time, hop by hop. And when a link fails, this layer can choose a different path, which is exactly why the internet survives partial failures.

Slide 14 · The transport layer

Layer four gets data from one application to another, end to end. It does that with port numbers, which is how one machine runs a web server, a mail server and a file share without confusing them. There are two protocols here and you must know the difference clearly. Transmission Control Protocol sets up a connection first, numbers every segment, and retransmits anything lost. User Datagram Protocol does none of that, which makes it faster and lower latency. So the question to ask is always the same. Does this application need reliability, or does it need speed?

Slide 15 · TCP versus UDP

This comparison appears in exam questions constantly, so let us take it one row at a time. Connection: TCP is connection oriented, UDP is connectionless. Reliability: TCP guarantees delivery and retransmits, UDP is best effort. Ordering: TCP numbers its segments and keeps them in order, UDP guarantees nothing. Speed: TCP is slower because of that overhead, UDP is faster. Typical use: TCP carries web, email and file transfer, where a missing byte matters; UDP carries voice, video and lookups. Think about a video call. If a packet arrives too late, you do not want it retransmitted, because by then the moment has passed. That is why real-time traffic prefers UDP.

Slide 16 · The upper layers

The top three layers are often grouped together, but know each one separately. The session layer establishes, manages and terminates conversations between applications. It handles dialogue control, deciding whose turn it is to talk, and it can place checkpoints so a long transfer does not restart from scratch. The presentation layer is the translator: it converts between data formats, and it handles compression and encryption. The application layer is where the familiar protocols live, including HTTP for the web and DNS for name resolution. And be careful here, because this is an exam trap. The application layer is the protocol layer, not the program itself.

Slide 17 · Discussion

[Run the discussion per the instructor notes.] So why does naming the layer first work so well? Because each layer has one job, and therefore each layer has its own attacks and its own defences. An attack against a web

application arrives at layer seven, so your answer is a web application firewall. A spoofed packet arrives at layer three, so you filter at the router. A rogue switch, or a poisoned address table, is layer two, so you use port security. Name the layer, and the control usually names itself.

Slide 18 · Break

Ten minutes. When we return, the media your signals actually travel through, the topologies we arrange them in, and then IPv4 addressing and subnetting, which is where the arithmetic starts.

Slide 19 · Wireless transmission media

Wireless media use radio waves to carry data through the air, and the obvious advantage is mobility, with no cabling to install. The spectrum is divided into frequency bands, and the band you choose is a real trade-off. Lower frequencies travel further and pass through walls better, but they carry less data. Higher frequencies carry much more data over shorter distances. The fundamental limitation is that wireless is shared. Every device on the channel competes for the same airtime, so performance drops as clients join and drops further with outside interference.

Slide 20 · Wired transmission media

Wired media come in a few families. Twisted pair is the cheapest and easiest to install, which is why it fills every office. The categories rate its capability, with Category five enhanced and Category six the ones you will meet most. Shielding reduces interference from nearby electrical sources such as motors and lighting. Coaxial carries higher frequencies than twisted pair, and survives in older installations and cable broadband. Fibre carries data as light rather than electricity, so it is immune to interference and spans the longest distances at the highest bandwidth. If you are running a link between two buildings, fibre is almost always the answer.

Slide 21 · Choosing the right cable

This table is the decision aid I want you to carry away. Twisted pair suits office desktops, short runs and tight budgets. Shielded twisted pair is the same idea with a shield, for electrically noisy places such as a factory floor. Coaxial serves legacy networks and cable broadband. Fibre optic is for long distance, very high bandwidth, or a link between two buildings. Notice that the deciding factors are always the same three questions. How far? How much bandwidth? How much interference? Answer those and the cable chooses itself.

Slide 22 · Connectors and transceivers

Connectors and transceivers are the parts that join things together, and they matter more than people expect. A transceiver converts between signal types, which lets a copper port talk to a fibre link. On copper, the standard connector is the Registered Jack 45, which you will plug into almost every desktop in the building. On fibre, the two to recognise are the Subscriber Connector and the Lucent Connector. A Small Form factor

Pluggable module is hot swappable, so you can change the media type without powering down the switch, which is a real operational benefit. And media converters bridge a copper segment to a fibre segment. Treat connector choice as a design decision, because it affects how easy the network is to maintain.

Slide 23 · Star, mesh and hybrid

Topology is how we arrange the connections. In a star, every device connects to a central point, usually a switch. That is easy to troubleshoot, because each device has its own cable. The drawback is that the central point is a single point of failure, so if that switch dies, everything on it goes too. A full mesh links every node to every other node, giving maximum resilience because any link can fail, but cost and complexity grow fast. A hybrid meshes the parts that need resilience and stars the rest, which is why most real networks are hybrids. Every topology answers one question: how much will you spend to survive a failure?

Slide 24 · Star against mesh

A star is simple, and one fault affects one device, but its central point can fail. A full mesh has no single point of failure, but it is expensive and complex. A hybrid gives resilience where it matters, at the cost of careful design. In data centres you also meet spine and leaf, where every leaf switch connects to every spine switch. That gives predictable latency, because any two devices sit the same number of hops apart. Four patterns, one trade-off, and you will choose between them in the design questions of later sessions.

Slide 25 · IPv4 address fundamentals

Now the addressing that makes routing possible: Internet Protocol version four. Let us start with the structure. An IPv4 address is thirty two bits long. We split those bits into four groups of eight, and each group is an octet. We write it in dotted decimal, so we say one nine two dot one six eight dot one dot one. The reason is readability, because the machine sees thirty two binary digits and you cannot read those aloud. Each octet runs from zero to two hundred and fifty five, and the whole space is about four point two nine billion addresses. That is why nobody expected to run out, and why subnetting exists at all.

Slide 26 · Addresses to recognise

Some addresses you should recognise on sight, and this is a slide worth memorising. The private ranges are not routable on the public internet, which lets every organisation reuse them internally. They are ten slash eight, one seven two dot sixteen slash twelve, and one nine two dot one six eight slash sixteen. The loopback address, one two seven dot zero dot zero dot one, always means the local machine. Then there is the range beginning one six nine dot two five four, which a machine gives itself when no server answers its address request. When you see one of those, you have found a symptom, not a configuration. Investigate the failure, not the address.

Slide 27 · Subnet masks

The subnet mask is the key to this whole subject. A mask is also thirty two bits: a run of ones followed by a run of zeros. The ones mark the network, the zeros mark the host. So a mask of two five five dot two five five dot two five five dot zero means the first three octets are the network. Count carefully here, because this is where people lose marks. Eight host bits give two hundred and fifty six values, but the lowest is the network address and the highest is the broadcast. That leaves two hundred and fifty four usable hosts, in every subnet, without exception.

Slide 28 · Worked example

Let me work one through. Take the address one nine two dot one six eight dot one dot fifty with the mask two five five dot two five five dot two five five dot zero. The mask tells us the first three octets are the network, so the network address is one nine two dot one six eight dot one dot zero. The usable hosts run from one nine two dot one six eight dot one dot one up to one nine two dot one six eight dot one dot two five four. And the broadcast is one nine two dot one six eight dot one dot two five five. The pattern never changes: find the network, count up from one, and stop one short of the broadcast. Do that quickly and you have the foundation for everything else in this chapter.

Slide 29 · Classless addressing

Classless addressing replaced the rigid old class system. Classless Inter-Domain Routing states how many bits are the network using a slash and a number. So slash twenty four means twenty four network bits, matching two five five dot two five five dot two five five dot zero. Here is the counter-intuitive part: a larger slash number means a smaller network. Slash thirty gives just two addresses, which is the standard for a link between two routers. Variable Length Subnet Masking, or VLSM, lets each subnet be a different size, so servers get large subnets and router links get tiny ones, and the rule is to work largest first. Classful addressing wasted whole blocks; classless lets the boundary fall at any bit.

Slide 30 · Prefix notation

This table covers almost every subnet you will size in practice. A slash twenty four gives two hundred and fifty four usable addresses, the classic office network. A slash twenty five halves that to one hundred and twenty six. A slash twenty six gives sixty two, a slash twenty seven gives thirty, and a slash thirty gives just two. Notice the pattern. Every time the prefix grows by one bit you halve the available addresses, and you always lose two to the network and the broadcast. Learn this table and you will answer half the subnetting questions you will ever see.

Slide 31 · Activity

[Run the activity per the instructor notes. Six minutes solo, four minutes debrief.] Five quick calculations. The network address of one nine two dot one six eight dot ten dot fifty slash twenty eight. Block size sixteen, fifty falls in the forty eight to sixty three range, so the network is one nine two dot one six eight dot ten dot forty

eight. A slash twenty nine has three host bits: eight minus two is six usable hosts. Mask two five five dot two five five dot two five five dot two four zero is slash twenty eight. Borrow two bits from a slash twenty four and you get four subnets of sixty two hosts each. And the address you can never assign is the network or broadcast address.

Slide 32 · Quiz

[Direct candidates to Quiz_Session_01.html. Fifteen minutes.] Open the session quiz. Twenty questions, immediate feedback, and every question is tagged with its topic area, so you will see the tag, like one B, IPv4 Addressing, above each question. Read every explanation, because the distractors teach as much as the answers. Aim for seventy percent or better before you leave today. There is also a Wayground version of this quiz for in-class assessment.

Slide 33 · Key takeaways

Eight takeaways. One: seven layers, one job each, and that is what makes faults findable. Two: encapsulation changes the name, from data to segment, packet, frame and finally bits. Three: layer two is local, delivering frames with forty eight bit hardware addresses. Four: layer three routes with IP addresses between networks, and Time to Live stops loops. Five: Transmission Control Protocol guarantees delivery while User Datagram Protocol favours speed. Six: media is a trade between distance, bandwidth and interference. Seven: every topology trades cost against resilience, so decide where you can afford to fail. And eight: the mask splits network from host, so a slash twenty four gives two hundred and fifty four usable hosts, not two hundred and fifty six.

Slide 34 · Revision checklist

Before next session: retake the quiz until you score seventy percent without guessing. Practise subnetting, ten calculations by hand from slash twenty four to slash thirty. Walk the stack out loud, naming each layer, its job and one protocol. Choose three cables for three scenarios and justify each with distance, bandwidth and interference. And read ahead to Session 2, where we build the layer two picture.

Slide 35 · Next session preview

Next time: Switching, VLANs and Network Architectures. We will follow frames through switches, learn how the forwarding table is built, split the network into virtual LANs with eight oh two point one Q trunking, and meet Spanning Tree Protocol, the loop-prevention safety net that keeps layer two networks from collapsing. If you take one habit from today, make it this: when something breaks, name the layer first, and the right tool usually presents itself. Thank you for your attention, and I will see you in Session Two.